Example-1

The conducting triangular loop in Figure 1(a) carries a current of 10 A. Find **H** at $(0, 0, 5)$ due to side 1 of the loop.

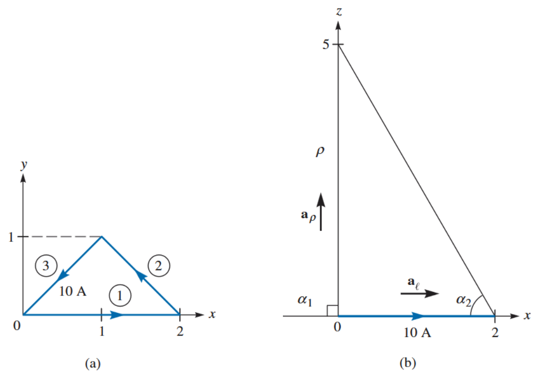

Figure 1: (a) conducting triangular loop, (b) side 1 of the loop.

Solution

This example illustrates how the Biot–Savart formulation for a straight, thin, current-carrying conductor is applied. The key point is determining α_1, α_2, ρ, and $\mathbf{a}_\phi$.

To find **H** at $(0, 0, 5)$ due to side 1 of the loop in Figure 1(a), consider Figure 1(b), where side 1 is treated as a straight conductor.

The observation point $(0, 0, 5)$ is joined to the beginning and end of the conductor. The angles α_1, α_2, and distance ρ are assigned accordingly.

$$\cos\alpha_1 = \cos 90^\circ = 0, \qquad \cos\alpha_2 = \frac{2}{\sqrt{29}}, \qquad \rho = 5$$

To determine $\mathbf{a}_\phi$, note that

$$\mathbf{a}_\ell = \mathbf{a}_x, \qquad \mathbf{a}_\rho = \mathbf{a}_z$$

Hence,

$$\mathbf{a}_\phi = \mathbf{a}_\ell \times \mathbf{a}_\rho = \mathbf{a}_x \times \mathbf{a}_z = -\mathbf{a}_y$$

Hence,

$$\begin{aligned}\mathbf{H}_1 &= \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\,\mathbf{a}_\phi \\ &= \frac{10}{4\pi(5)}\left(\frac{2}{\sqrt{29}} - 0\right)(-\mathbf{a}_y) \\ &= -59.1\,\mathbf{a}_y \text{ mA/m}\end{aligned}$$

Example-2

Find $\mathbf{H}$ at $(0,0,5)$ due to side 3 (Fig 1a) of the triangular loop.

Solution

The magnetic field due to a finite straight conductor is

$$\mathbf{H} = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\mathbf{a}_\phi$$

Side 3 extends from $(0,0,0)$ to $(1,1,0)$ and the observation point is $(0,0,5)$.

$$\rho = 5$$

The direction of current is

$$\mathbf{a}_\ell = \frac{\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}}$$

Since $\mathbf{a}_\rho = \mathbf{a}_z$, we get

$$\mathbf{a}_\phi = \mathbf{a}_\ell \times \mathbf{a}_\rho = \frac{1}{\sqrt{2}}(\mathbf{a}_x + \mathbf{a}_y) \times \mathbf{a}_z$$

$$= \frac{1}{\sqrt{2}}(-\mathbf{a}_y + \mathbf{a}_x) = \frac{1}{\sqrt{2}}(\mathbf{a}_x - \mathbf{a}_y)$$

From geometry,

$$\cos\alpha_1 = 0, \qquad \cos\alpha_2 = \frac{2}{\sqrt{29}}$$

Substituting,

$$\mathbf{H}_3 = \frac{10}{4\pi(5)}\left(\frac{2}{\sqrt{29}}\right)\mathbf{a}_\phi$$

$$= \frac{10}{20\pi} \cdot \frac{2}{\sqrt{29}} \cdot \frac{1}{\sqrt{2}}(\mathbf{a}_x - \mathbf{a}_y)$$

$$\mathbf{H}_3 = -30.63\,\mathbf{a}_x + 30.63\,\mathbf{a}_y \text{ mA/m}$$

Example-3

Find $\mathbf{H}$ at $(-3, 4, 0)$ due to the current filament shown in Fig 2.

Solution

Let

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2$$

where $\mathbf{H}_1$ and $\mathbf{H}_2$ are the contributions due to the portions of the filament along the x- and z-axes, respectively.

Field due to z-axis segment ($\mathbf{H}_2$)

$$\mathbf{H}_2 = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\mathbf{a}_\phi$$

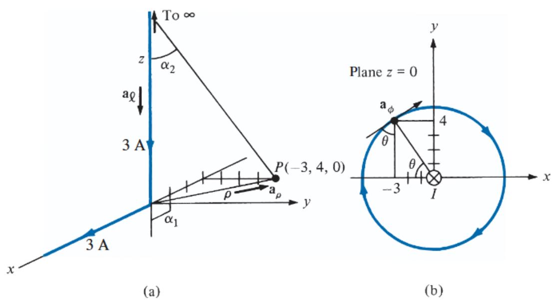

Figure 2: (a) current filament along semi-infinite x- and z-axes, $\mathbf{a}_\ell$ and $\mathbf{a}_\rho$ for $\mathbf{H}_2$ only; (b) determining $\mathbf{a}_\phi$ for $\mathbf{H}_2$

At $P(-3, 4, 0)$,

$$\rho = \sqrt{9 + 16} = 5, \quad \alpha_1 = 90^\circ, \quad \alpha_2 = 0^\circ$$

The unit vector $\mathbf{a}_\phi$ is determined using geometry:

$$\mathbf{a}_\phi = \sin\theta\,\mathbf{a}_x + \cos\theta\,\mathbf{a}_y = \frac{4}{5}\mathbf{a}_x + \frac{3}{5}\mathbf{a}_y$$

Alternatively,

$$\mathbf{a}_\phi = -\mathbf{a}_z \times \left(-\frac{3}{5}\mathbf{a}_x + \frac{4}{5}\mathbf{a}_y\right) = \frac{4}{5}\mathbf{a}_x + \frac{3}{5}\mathbf{a}_y$$

Thus,

$$\mathbf{H}_2 = \frac{3}{4\pi(5)}(1 - 0)\frac{4\mathbf{a}_x + 3\mathbf{a}_y}{5}$$

$$\mathbf{H}_2 = 38.2\,\mathbf{a}_x + 28.65\,\mathbf{a}_y \text{ mA/m}$$

Alternatively in cylindrical form,

$$\mathbf{H}_2 = \frac{3}{4\pi(5)}(1-0)(-\mathbf{a}_\phi) = -47.75\,\mathbf{a}_\phi \text{ mA/m}$$

Field due to x-axis segment ($\mathbf{H}_1$)

At P,

$$\rho = 4, \quad \alpha_2 = 0^\circ, \quad \cos\alpha_1 = \frac{3}{5}$$

Also,

$$\mathbf{a}_\phi = \mathbf{a}_z$$

Hence,

$$\mathbf{H}_1 = \frac{3}{4\pi(4)}\left(1 - \frac{3}{5}\right)\mathbf{a}_z$$

$$\mathbf{H}_1 = 23.87\,\mathbf{a}_z \text{ mA/m}$$

Total Field

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 = 38.2\,\mathbf{a}_x + 28.65\,\mathbf{a}_y + 23.87\,\mathbf{a}_z \text{ mA/m}$$

or

$$\mathbf{H} = -47.75\,\mathbf{a}_\phi + 23.87\,\mathbf{a}_z \text{ mA/m}$$

Example 4

A thin ring of radius 5 cm is placed on plane $z = 1$ cm so that its center is at $(0, 0, 1 \text{ cm})$. If the ring carries 50 mA along $\mathbf{a}_\phi$, find $\mathbf{H}$ at: (a) $(0, 0, -1 \text{ cm})$ (b) $(0, 0, 10 \text{ cm})$

Solution:

Given:

- Radius: $a = 5$ cm $= 0.05$ m
- Current: $I = 50$ mA $= 0.05$ A
- Loop center at $z = 1$ cm $= 0.01$ m

Magnetic field on the axis of a circular loop is:

$$H_z = \frac{Ia^2}{2(a^2 + z^2)^{3/2}}$$

(a) At $(0, 0, -1\ \mathbf{cm})$

Shift coordinate:

$$z' = -0.01 - 0.01 = -0.02 \text{ m}$$

$$H_z = \frac{0.05 \times (0.05)^2}{2\left[(0.05)^2 + (-0.02)^2\right]^{3/2}}$$
$$= \frac{0.000125}{2(0.0029)^{3/2}}$$

$$(0.0029)^{3/2} \approx 0.000156$$

$$H_z = \frac{0.000125}{0.000312} = 0.400 \text{ A/m}$$

$$\boxed{H = 400\,\mathbf{a}_z \text{ mA/m}}$$

(b) At $(0, 0, 10\ \mathbf{cm})$

$$z' = 0.10 - 0.01 = 0.09 \text{ m}$$

$$H_z = \frac{0.000125}{2(0.0106)^{3/2}}$$

$$(0.0106)^{3/2} \approx 0.001092$$

$$H_z = \frac{0.000125}{0.002184} = 0.0573 \text{ A/m}$$

$$\boxed{H = 57.3\,\mathbf{a}_z \text{ mA/m}}$$

Example 5:

The solenoid of Figure 3 has 2000 turns, a length of 75 cm, and a radius of 5 cm. If it carries a current of 50 mA along $\mathbf{a}_\phi$, find $\mathbf{H}$ at

(a) $(0, 0, 0)$

(b) $(0, 0, 75 \text{ cm})$

(c) $(0, 0, 50 \text{ cm})$

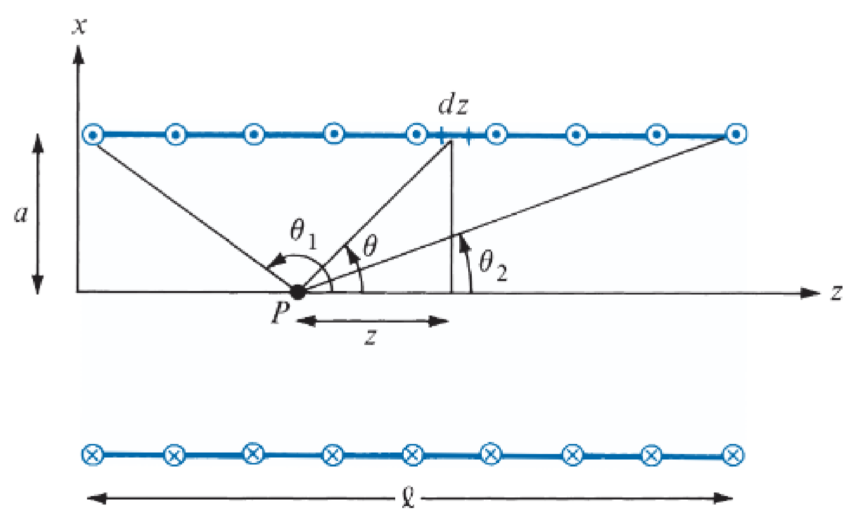

Figure 3: cross section of a solenoid.

Solution:

The solenoid has:

$$N = 2000, \quad l = 75 \text{ cm} = 0.75 \text{ m}, \quad a = 5 \text{ cm} = 0.05 \text{ m}$$

$$I = 50 \text{ mA} = 0.05 \text{ A}$$

Number of turns per unit length:

$$n = \frac{N}{l} = \frac{2000}{0.75} = 2666.67 \text{ turns/m}$$

Magnetic field on the axis of a finite solenoid:

$$H_z = \frac{nI}{2}(\cos\theta_1 + \cos\theta_2)$$

where

$$\cos\theta = \frac{z}{\sqrt{z^2 + a^2}}$$

(a) At $(0, 0, 0)$

Distances to ends:

$$z_1 = 0, \quad z_2 = 0.75$$

$$\cos\theta_1 = 0, \quad \cos\theta_2 = \frac{0.75}{\sqrt{0.75^2 + 0.05^2}} \approx 0.9978$$

$$H_z = \frac{2666.67 \times 0.05}{2}(0 + 0.9978)$$

$$H_z \approx 66.52 \text{ A/m}$$

$$\mathbf{H} = 66.52\,\mathbf{a}_z \text{ A/m}$$

(b) At $(0, 0, 75$ cm$)$

$$z_1 = 0.75, \quad z_2 = 0$$

$$\cos\theta_1 \approx 0.9978, \quad \cos\theta_2 = 0$$

$$H_z = \frac{2666.67 \times 0.05}{2}(0.9978)$$

$$\mathbf{H} = 66.52\,\mathbf{a}_z \text{ A/m}$$

(c) At $(0, 0, 50$ cm$)$

$$z_1 = 0.50, \quad z_2 = 0.25$$

$$\cos\theta_1 = \frac{0.50}{\sqrt{0.50^2 + 0.05^2}} \approx 0.9950$$

$$\cos\theta_2 = \frac{0.25}{\sqrt{0.25^2 + 0.05^2}} \approx 0.9806$$

$$H_z = \frac{2666.67 \times 0.05}{2}(0.9950 + 0.9806)$$

$$H_z \approx 131.7 \text{ A/m}$$

$$\mathbf{H} = 131.7\,\mathbf{a}_z \text{ A/m}$$

Example 6:

Planes $z = 0$ and $z = 4$ carry current $\mathbf{K} = -10\mathbf{a}_x$ A/m and $\mathbf{K} = 10\mathbf{a}_x$ A/m, respectively. Determine $\mathbf{H}$ at

(a) $(1, 1, 1)$

(b) $(0, -3, 10)$

Solution:

The parallel current sheets are shown in Figure 4. Let

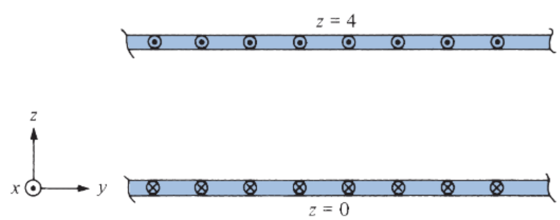

Figure 4: parallel infinite current sheets

$$\mathbf{H} = \mathbf{H}_0 + \mathbf{H}_4$$

where $\mathbf{H}_0$ and $\mathbf{H}_4$ are the contributions due to the current sheets $z = 0$ and $z = 4$, respectively. We make use of equation derived for the case of Infinite Sheet of Current.

(a) At $(1, 1, 1)$, which is between the plates $(0 < z = 1 < 4)$,

$$\mathbf{H}_0 = \frac{1}{2}\mathbf{K} \times \mathbf{a}_n = \frac{1}{2}(-10\mathbf{a}_x) \times \mathbf{a}_z = 5\mathbf{a}_y \text{ A/m}$$

$$\mathbf{H}_4 = \frac{1}{2}\mathbf{K} \times \mathbf{a}_n = \frac{1}{2}(10\mathbf{a}_x) \times (-\mathbf{a}_z) = 5\mathbf{a}_y \text{ A/m}$$

Hence,

$$\mathbf{H} = 10\mathbf{a}_y \text{ A/m}$$

(b) At $(0, -3, 10)$, which is above the two sheets $(z = 10 > 4 > 0)$,

$$\mathbf{H}_0 = \frac{1}{2}(-10\mathbf{a}_x) \times \mathbf{a}_z = 5\mathbf{a}_y \text{ A/m}$$

$$\mathbf{H}_4 = \frac{1}{2}(10\mathbf{a}_x) \times \mathbf{a}_z = -5\mathbf{a}_y \text{ A/m}$$

Hence,

$$\mathbf{H} = 0 \text{ A/m}$$

Example 7:

A toroid whose dimensions are shown in Figure 5 has N turns and carries current I. Determine $\mathbf{H}$ inside and outside the toroid.

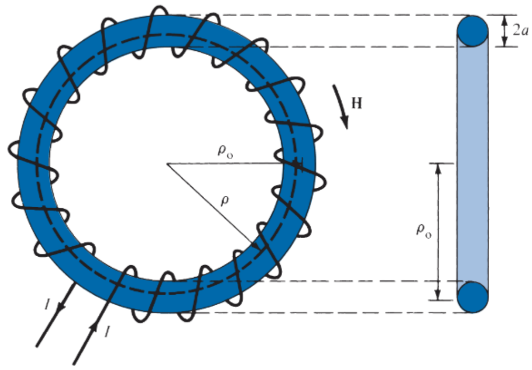

Figure 5: A toroid with a circular cross section.

Solution:

We apply Ampère's circuit law to the Amperian path, which is a circle of radius ρ shown dashed in Figure 5. Since N wires cut through this path each carrying current I, the net current enclosed by the Amperian path is NI. Hence,

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} \;\rightarrow\; H \cdot 2\pi\rho = NI$$

or

$$H = \frac{NI}{2\pi\rho}, \quad \text{for } \rho_0 - a < \rho < \rho_0 + a$$

where ρ_0 is the mean radius of the toroid as shown in Figure 5. An approximate value of H is

$$H_{\text{approx}} = \frac{NI}{2\pi\rho_0} = \frac{NI}{\ell}$$

Notice that this is the same as the formula obtained for H for points well inside a very long solenoid ($\ell \gg a$). Thus a straight solenoid may be regarded as a special toroidal coil for which $\rho_0 \rightarrow \infty$. Outside the toroid, the current enclosed by an Amperian path is $NI - NI = 0$ and hence $H = 0$.

Example 8:

A toroid of circular cross section whose center is at the origin and axis the same as the z-axis has 1000 turns with $\rho_0 = 10$ cm, $a = 1$ cm. If the toroid carries a 100 mA current, find $|\mathbf{H}|$ at

(a) $(3 \text{ cm}, -4 \text{ cm}, 0)$

(b) $(6 \text{ cm}, 9 \text{ cm}, 0)$

Solution:

Given:

$$N = 1000, \quad \rho_0 = 10 \text{ cm} = 0.1 \text{ m}, \quad a = 1 \text{ cm} = 0.01 \text{ m}$$

$$I = 100 \text{ mA} = 0.1 \text{ A}$$

Magnetic field in a toroid:

$$H = \frac{NI}{2\pi\rho}, \quad \text{for } \rho_0 - a < \rho < \rho_0 + a$$

Outside this region:

$$H = 0$$

(a) At $(3, -4, 0)$

$$\rho = \sqrt{3^2 + (-4)^2} = 5 \text{ cm}$$

$$\rho_0 - a = 9 \text{ cm}, \quad \rho_0 + a = 11 \text{ cm}$$

Since $\rho < 9$ cm, the point lies outside the toroid.

$$\boxed{H = 0}$$

(b) At $(6, 9, 0)$

$$\rho = \sqrt{6^2 + 9^2} = \sqrt{117} \approx 10.82 \text{ cm} = 0.1082 \text{ m}$$

Since 9 cm $< \rho <$ 11 cm, the point lies inside the toroid.

$$H = \frac{1000 \times 0.1}{2\pi \times 0.1082}$$

$$H \approx 147.1 \text{ A/m}$$

$$\boxed{H = 147.1 \text{ A/m}}$$

Example 9:

If plane $z = 0$ carries uniform current $\mathbf{K} = K_y \mathbf{a}_y$,

$$\mathbf{H} = \begin{cases} \frac{1}{2} K_y \mathbf{a}_x, & z > 0 \\ -\frac{1}{2} K_y \mathbf{a}_x, & z < 0 \end{cases}$$

This was obtained by using Ampère's law. Obtain this by using the concept of vector magnetic potential.

Solution:

Consider the current sheet as in Figure 6. From eq. Magnetic vector for for surface current

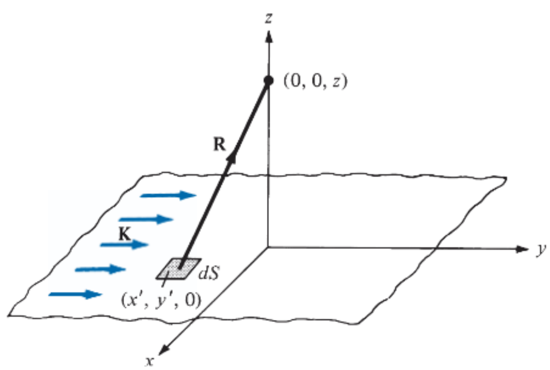

Figure 6: Infinite current sheet.

$$d\mathbf{A} = \frac{\mu_0 \mathbf{K}\, dS}{4\pi R}$$

In this problem, $\mathbf{K} = K_y \mathbf{a}_y$, $dS = dx'\, dy'$, and for $z > 0$,

$$R = |\mathbf{R}| = |(0, 0, z) - (x', y', 0)|$$

$$= \left[(x')^2 + (y')^2 + z^2\right]^{1/2} \tag{7.8.1}$$

where the primed coordinates are for the source point while the unprimed coordinates are for the field point. It is necessary (and customary) to distinguish between the two points to avoid confusion (see Illustration of the source point (x', y', z') and the field point (x, y, z)). Hence

$$d\mathbf{A} = \frac{\mu_0 K_y\, dx'\, dy'\, \mathbf{a}_y}{4\pi \left[(x')^2 + (y')^2 + z^2\right]^{1/2}}$$

$$d\mathbf{B} = \nabla \times d\mathbf{A} = -\frac{\partial}{\partial z} dA_y\, \mathbf{a}_x$$

$$= \frac{\mu_0 K_y z \, dx' \, dy' \, \mathbf{a}_x}{4\pi \left[(x')^2 + (y')^2 + z^2\right]^{3/2}}$$

$$\mathbf{B} = \frac{\mu_0 K_y z \mathbf{a}_x}{4\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dx' \, dy'}{\left[(x')^2 + (y')^2 + z^2\right]^{3/2}} \tag{7.8.2}$$

In the integrand, we may change coordinates from Cartesian to cylindrical for convenience so that

$$\mathbf{B} = \frac{\mu_0 K_y z \mathbf{a}_x}{4\pi} \int_{\phi'=0}^{2\pi} \int_{\rho'=0}^{\infty} \frac{\rho' \, d\phi' \, d\rho'}{\left[(\rho')^2 + z^2\right]^{3/2}}$$

$$= \frac{\mu_0 K_y z \mathbf{a}_x}{4\pi} \cdot 2\pi \int_0^{\infty} \left[(\rho')^2 + z^2\right]^{-3/2} \rho' \, d\rho'$$

$$= \frac{\mu_0 K_y z \mathbf{a}_x}{2} \int_0^{\infty} \left[(\rho')^2 + z^2\right]^{-3/2} \rho' \, d\rho'$$

$$= \frac{\mu_0 K_y z \mathbf{a}_x}{2} \left[\frac{-1}{\left[(\rho')^2 + z^2\right]^{1/2}} \right]_{\rho'=0}^{\infty}$$

$$= \frac{\mu_0 K_y \mathbf{a}_x}{2}$$

Hence

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} = \frac{K_y}{2} \mathbf{a}_x \quad \text{for } z > 0$$

By simply replacing z by $-z$ in eq. above and following the same procedure, we obtain

$$\mathbf{H} = -\frac{K_y}{2} \mathbf{a}_x \quad \text{for } z < 0$$

Example 10:

Given that $\mathbf{H}_1 = -2\mathbf{a}_x + 6\mathbf{a}_y + 4\mathbf{a}_z$ A/m in region $y - x - 2 \leq 0$, where $\mu_1 = 5\mu_0$, calculate

(a) $\mathbf{M}_1$ and $\mathbf{B}_1$

(b) $\mathbf{H}_2$ and $\mathbf{B}_2$ in region $y - x - 2 \geq 0$, where $\mu_2 = 2\mu_0$

Solution:

Since $y - x - 2 = 0$ is a plane, $y - x \leq 2$ or $y \leq x + 2$ is region 1 in Figure 7. A point in this region may be used to confirm this. For example, the origin $(0, 0)$ is in this region because $0 - 0 - 2 < 0$. If we let the surface of the plane be described by $f(x, y) = y - x - 2$, a unit vector normal to the plane is given by

$$\mathbf{a}_n = \frac{\nabla f}{|\nabla f|} = \frac{\mathbf{a}_y - \mathbf{a}_x}{\sqrt{2}}$$

(a)

$$\begin{aligned}\mathbf{M}_1 = \chi_{m1}\mathbf{H}_1 = (\mu_{r1} - 1)\mathbf{H}_1 = (5 - 1)(-2, 6, 4) \\ = -8\mathbf{a}_x + 24\mathbf{a}_y + 16\mathbf{a}_z \text{ A/m}\end{aligned}$$

$$\begin{aligned}\mathbf{B}_1 = \mu_1\mathbf{H}_1 = \mu_0\mu_{r1}\mathbf{H}_1 = 4\pi \times 10^{-7}(5)(-2, 6, 4) \\ = -12.57\mathbf{a}_x + 37.7\mathbf{a}_y + 25.13\mathbf{a}_z \text{ Wb/m}^2\end{aligned}$$

(b)

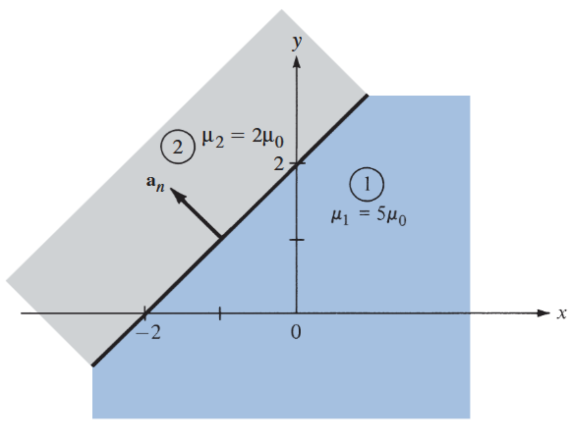

Figure 7: Magnetic boundary conditions example

$$\begin{aligned}\mathbf{H}_{1n} = (\mathbf{H}_1 \cdot \mathbf{a}_n)\mathbf{a}_n = (-2, 6, 4) \cdot \frac{(-1, 1, 0)}{\sqrt{2}} \cdot \frac{(-1, 1, 0)}{\sqrt{2}} \\ = -4\mathbf{a}_x + 4\mathbf{a}_y\end{aligned}$$

But

$$\mathbf{H}_1 = \mathbf{H}_{1n} + \mathbf{H}_{1t}$$

Hence,

$$\mathbf{H}_{1t} = \mathbf{H}_1 - \mathbf{H}_{1n} = (-2, 6, 4) - (-4, 4, 0)$$
$$= 2\mathbf{a}_x + 2\mathbf{a}_y + 4\mathbf{a}_z$$

Using the boundary conditions, we have

$$\mathbf{H}_{2t} = \mathbf{H}_{1t} = 2\mathbf{a}_x + 2\mathbf{a}_y + 4\mathbf{a}_z$$

$$\mathbf{B}_{2n} = \mathbf{B}_{1n} \Rightarrow \mu_2 \mathbf{H}_{2n} = \mu_1 \mathbf{H}_{1n}$$

or

$$\mathbf{H}_{2n} = \frac{\mu_1}{\mu_2}\mathbf{H}_{1n} = \frac{5}{2}(-4\mathbf{a}_x + 4\mathbf{a}_y) = -10\mathbf{a}_x + 10\mathbf{a}_y$$

Thus

$$\mathbf{H}_2 = \mathbf{H}_{2n} + \mathbf{H}_{2t} = -8\mathbf{a}_x + 12\mathbf{a}_y + 4\mathbf{a}_z \text{ A/m}$$

and

$$\mathbf{B}_2 = \mu_2 \mathbf{H}_2 = \mu_0 \mu_{r2} \mathbf{H}_2 = (4\pi \times 10^{-7})(2)(-8, 12, 4)$$
$$= -20.11\mathbf{a}_x + 30.16\mathbf{a}_y + 10.05\mathbf{a}_z \text{ Wb/m}^2$$

Example 11:

The xy-plane serves as the interface between two different media. Medium 1 ($z < 0$) is filled with a material whose $\mu_r = 6$, and medium 2 ($z > 0$) is filled with a material whose $\mu_r = 4$. If the interface carries current $(1/\mu_0)\,\mathbf{a}_y$ mA/m, and $\mathbf{B}_2 = 5\mathbf{a}_x + 8\mathbf{a}_z$ mWb/m^2, find $\mathbf{H}_1$ and $\mathbf{B}_1$.

Solution:

Consider the problem as illustrated in Figure 8. Let $\mathbf{B}_1 = (B_x, B_y, B_z)$ in mWb/m^2.

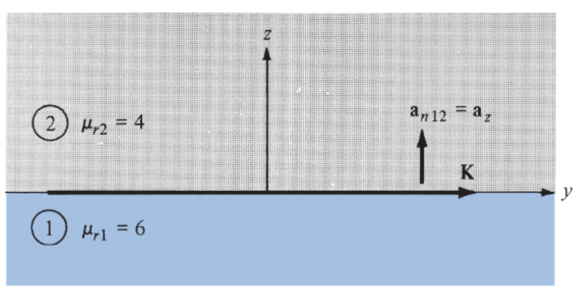

Figure 8: Illustrating figure for example 11

$$\mathbf{B}_{1n} = \mathbf{B}_{2n} = 8\mathbf{a}_z \to B_z = 8 \tag{1}$$

But

$$\mathbf{H}_2 = \frac{\mathbf{B}_2}{\mu_2} = \frac{1}{4\mu_0}(5\mathbf{a}_x + 8\mathbf{a}_z) \text{ mA/m} \tag{2}$$

and

$$\mathbf{H}_1 = \frac{\mathbf{B}_1}{\mu_1} = \frac{1}{6\mu_0}(B_x\mathbf{a}_x + B_y\mathbf{a}_y + B_z\mathbf{a}_z) \text{ mA/m} \tag{3}$$

Having found the normal components, we can find the tangential components by using

$$(\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{n12} = \mathbf{K}$$

or

$$\mathbf{H}_1 \times \mathbf{a}_{n12} = \mathbf{H}_2 \times \mathbf{a}_{n12} + \mathbf{K} \tag{4}$$

$$\frac{1}{6\mu_0}(B_x\mathbf{a}_x + B_y\mathbf{a}_y + B_z\mathbf{a}_z) \times \mathbf{a}_z = \frac{1}{4\mu_0}(5\mathbf{a}_x + 8\mathbf{a}_z) \times \mathbf{a}_z + \frac{1}{\mu_0}\mathbf{a}_y$$

Equating components yields

$$B_y = 0, \quad -\frac{B_x}{6} = -\frac{5}{4} + 1, \quad \text{or} \quad B_x = \frac{6}{4} = 1.5$$

(5)

$$\mathbf{B}_1 = 1.5\mathbf{a}_x + 8\mathbf{a}_z \text{ mWb/m}^2$$

$$\mathbf{H}_1 = \frac{\mathbf{B}_1}{\mu_1} = \frac{1}{\mu_0}(0.25\mathbf{a}_x + 1.33\mathbf{a}_z) \text{ mA/m}$$

and

$$\mathbf{H}_2 = \frac{1}{\mu_0}(1.25\mathbf{a}_x + 2\mathbf{a}_z) \text{ mA/m}$$

Note that H_{1x} is $1/\mu_0$ mA/m less than H_{2x} because of the current sheet and also that $B_{1n} = B_{2n}$.

Analogy

Electric	Magnetic
Conductivity σ	Permeability μ
Field intensity E	Field intensity H
Current $I = \int \mathbf{J} \cdot d\mathbf{S}$	Magnetic flux $\Psi = \int \mathbf{B} \cdot d\mathbf{S}$
Current density $J = \dfrac{I}{S} = \sigma E$	Flux density $B = \dfrac{\Psi}{S} = \mu H$
Electromotive force (emf) V	Magnetomotive force (mmf) $\mathcal{F}$
Resistance R	Reluctance $\mathcal{R}$
Conductance $G = \dfrac{1}{R}$	Permeance $\mathcal{P} = \dfrac{1}{\mathcal{R}}$
Ohm's law $R = \dfrac{V}{I} = \dfrac{\ell}{\sigma S}$ or $V = E\ell = IR$	Ohm's law $\mathcal{R} = \dfrac{\mathcal{F}}{\Psi} = \dfrac{\ell}{\mu S}$ or $\mathcal{F} = H\ell = \Psi \mathcal{R} = NI$
Kirchhoff's laws: $\sum I = 0$	Kirchhoff's laws: $\sum \Psi = 0$
$\sum V - \sum RI = 0$	$\sum \mathcal{F} - \sum \mathcal{R} \Psi = 0$

Figure 9: Analogy between Electric and Magnetic Circuits